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Studierichting: Informatica
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$$\int_{-1}^{2} (x \cdot \lfloor x \rfloor)\, dx = \int_{-1}^{0} (x \cdot \lfloor x \rfloor)\, dx + \int_{0}^{1} (x \cdot \lfloor x \rfloor)\, dx + \int_{1}^{2} (x \cdot \lfloor x \rfloor)\, dx$$

$$\int_{-1}^{0} (x \cdot \lfloor x \rfloor)\, dx = -\int_{-1}^{0} x dx = -\left[\frac{x^2}{2}\right]_{-1}^{0} = \frac{1}{2}$$

$$\int_{0}^{1} (x \cdot \lfloor x \rfloor)\, dx = 0 \cdot \int_{0}^{1} x dx = 0$$

$$\int_{1}^{2} (x \cdot \lfloor x \rfloor)\, dx = \int_{1}^{2} x dx = \left[\frac{x^2}{2}\right]_{1}^{2} = \frac{3}{2}$$

$$\Rightarrow \int_{-1}^{2} (x \cdot \lfloor x \rfloor)\, dx = \frac{1}{2} + 0 + \frac{3}{2} = 2$$

References